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(54) **REPLACEABLE CONDUCTIVE MARKER TIP**

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CPC **G06F 3/0442** (2019.05); **G06F 3/016** (2013.01); **G06F 3/03545** (2013.01)

(58) **Field of Classification Search**
CPC G06F 3/0442; G06F 3/016; G06F 3/03545
See application file for complete search history.

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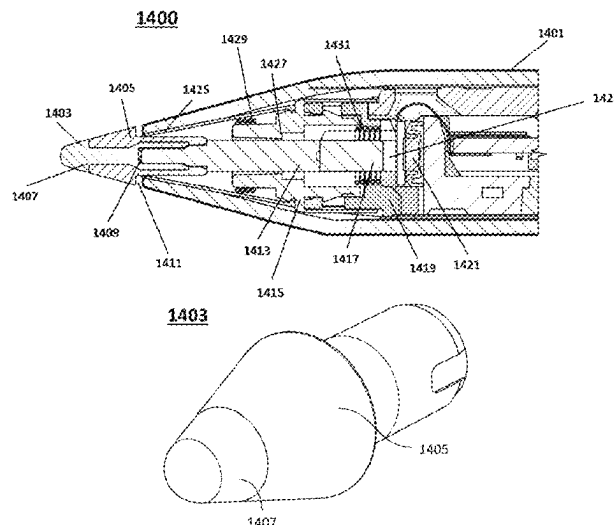
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(57) **ABSTRACT**

Embodiments of the invention provide a removable marker tip configured for application with an active pen-stylus. The removable marker tip comprises an antenna and an insulator and is designed to be held onto the pen-stylus' writing shaft by a set of crush ribs fitted into a posterior recess on the removable marker tip. The removable marker tip is designed to be a consumable part to improve the paper feeling received by users of the pen-stylus as they draw on the display of a tablet device. The removable marker tip is further designed to be hand-removable by users of the pen-stylus.

10 Claims, 10 Drawing Sheets



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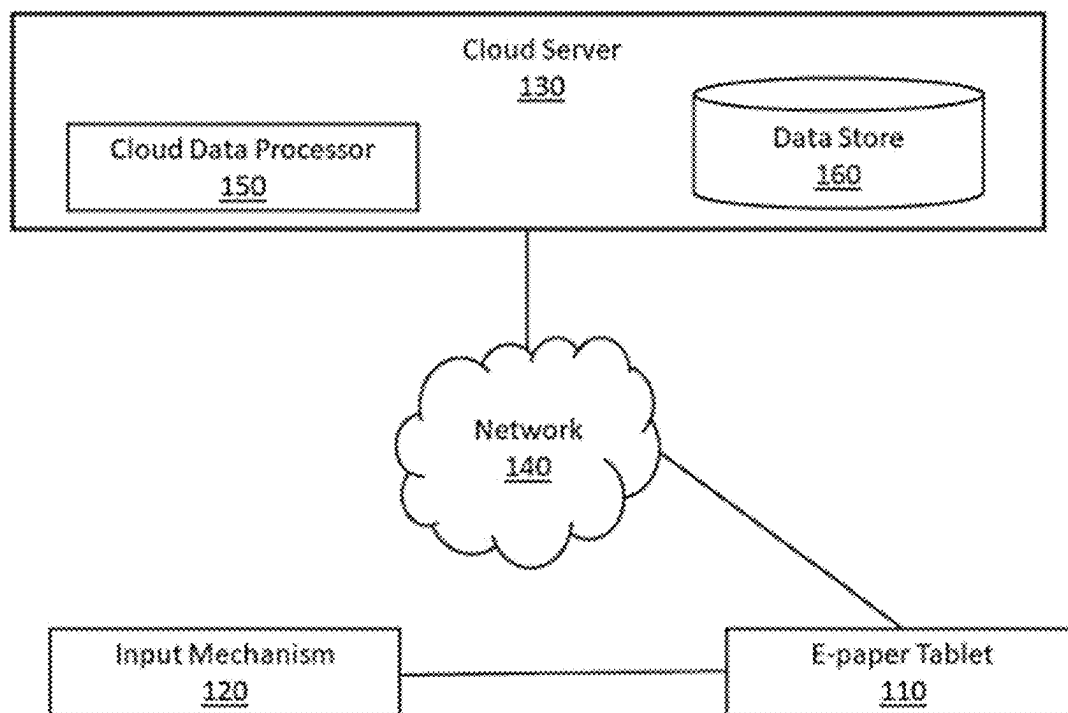


FIG. 1
(Prior Art)

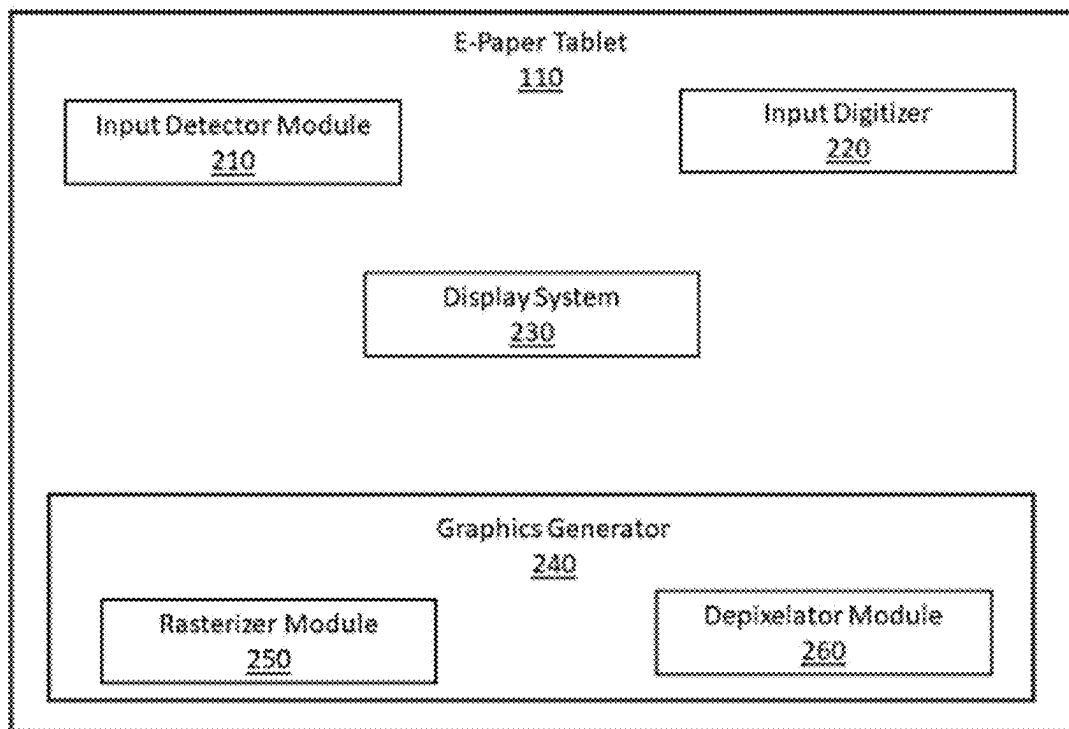


FIG. 2
(Prior Art)

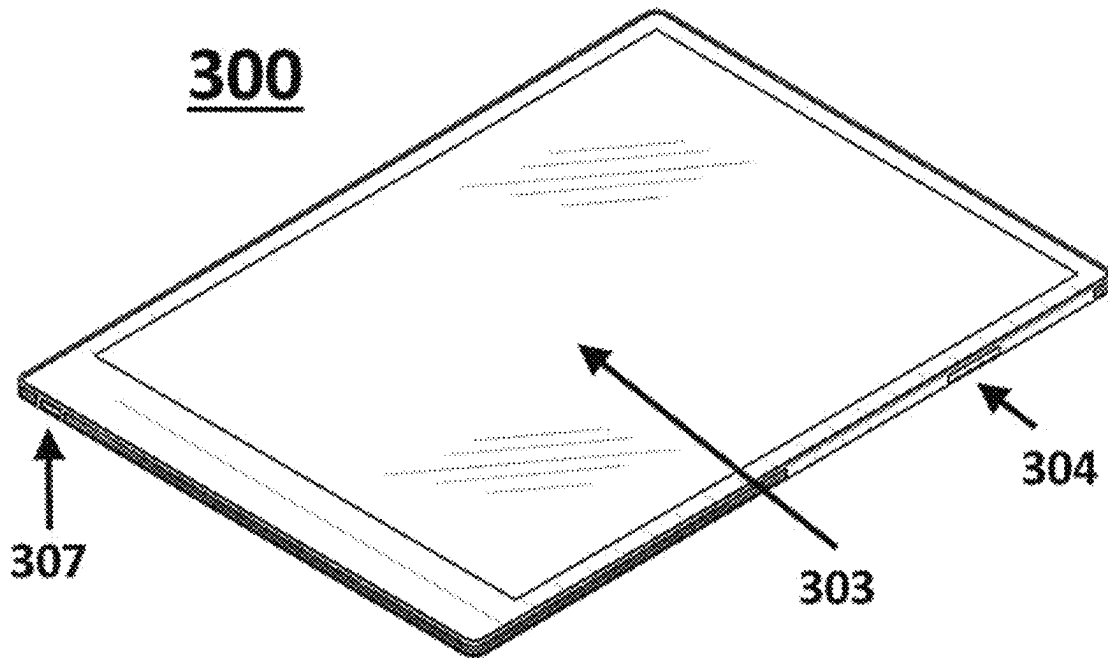


FIG. 3
(Prior Art)

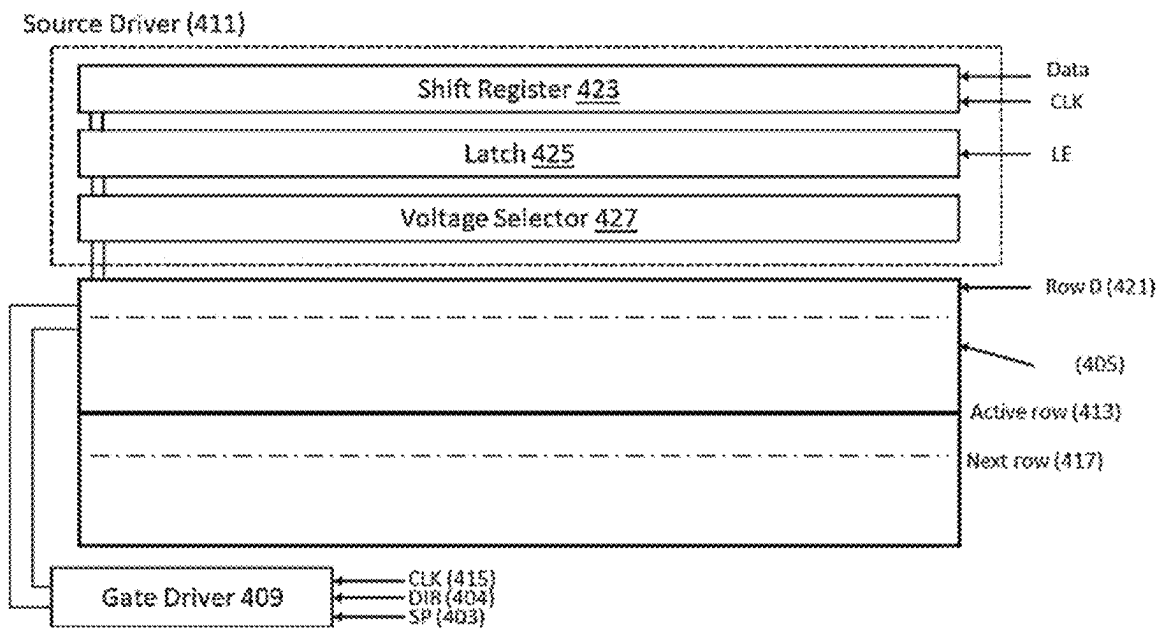


FIG. 4
(Prior Art)

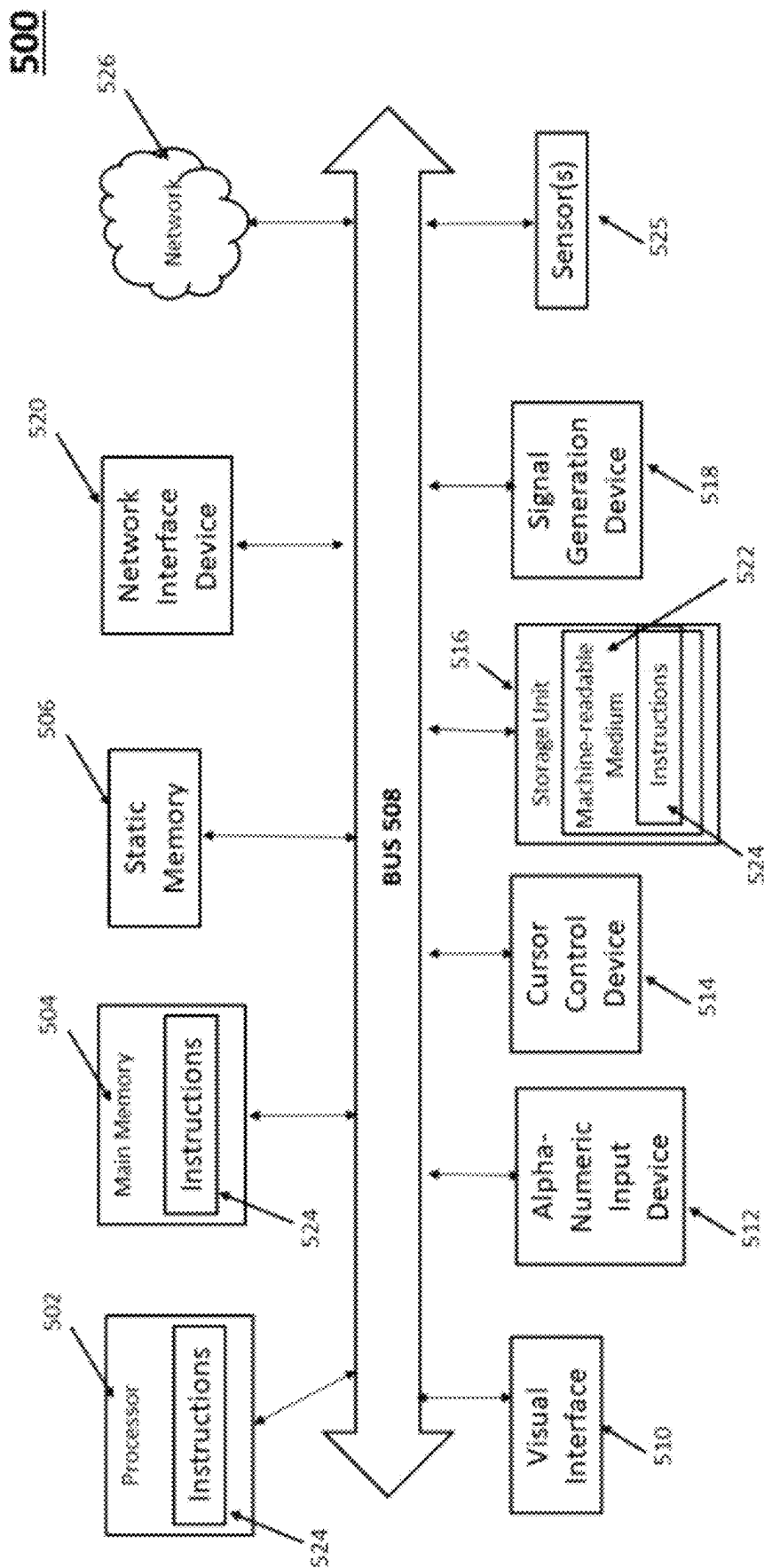


FIG. 5
(Prior Art)

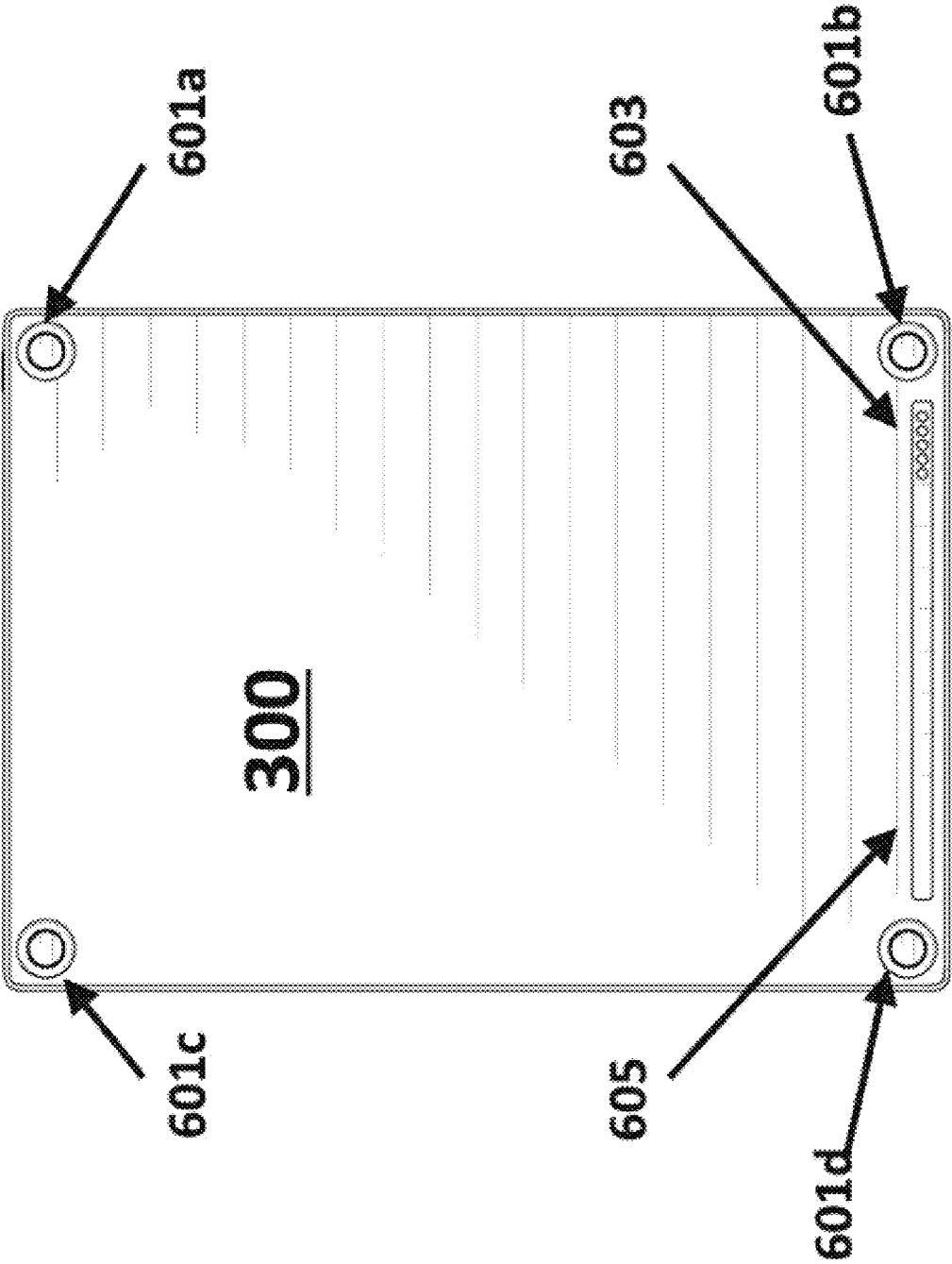


FIG. 6



FIG. 7

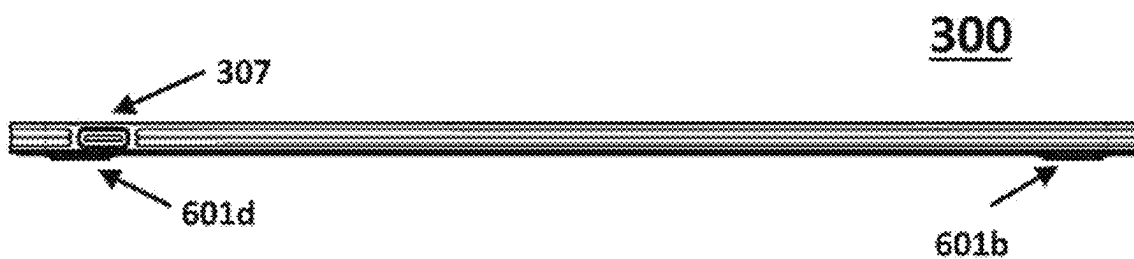


FIG. 8

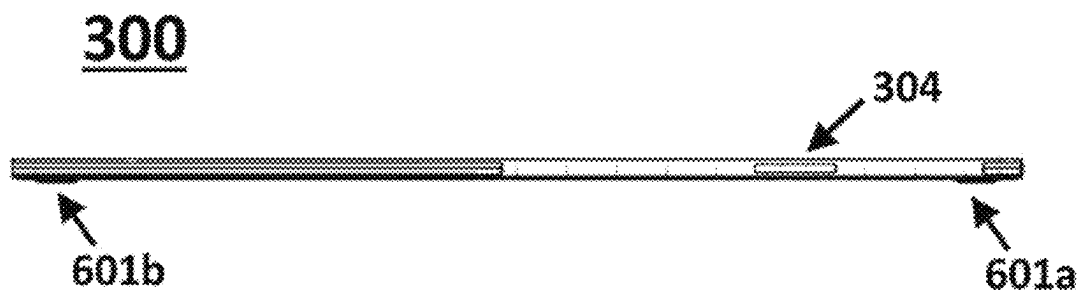


FIG. 9

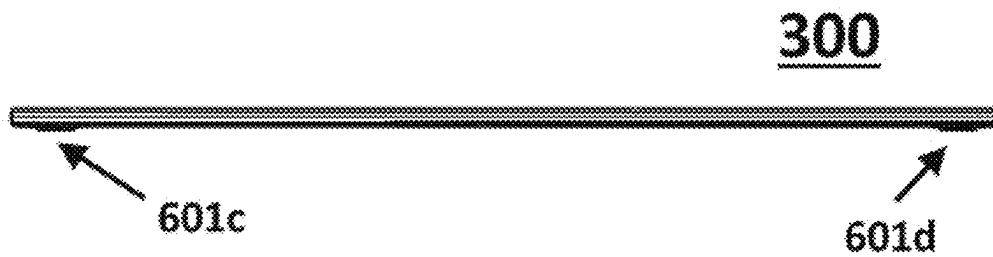


FIG. 10

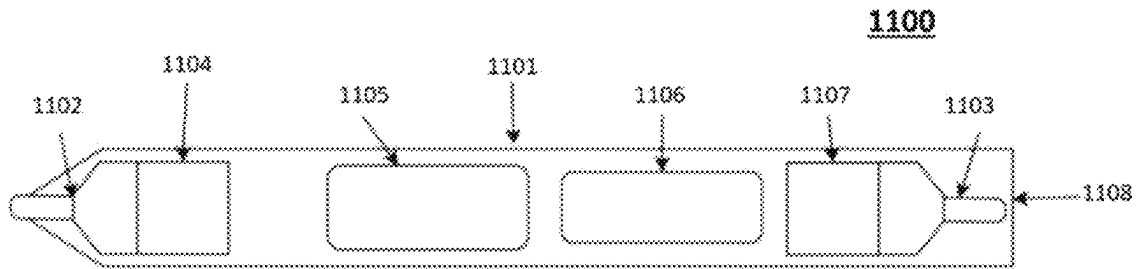


FIG. 11

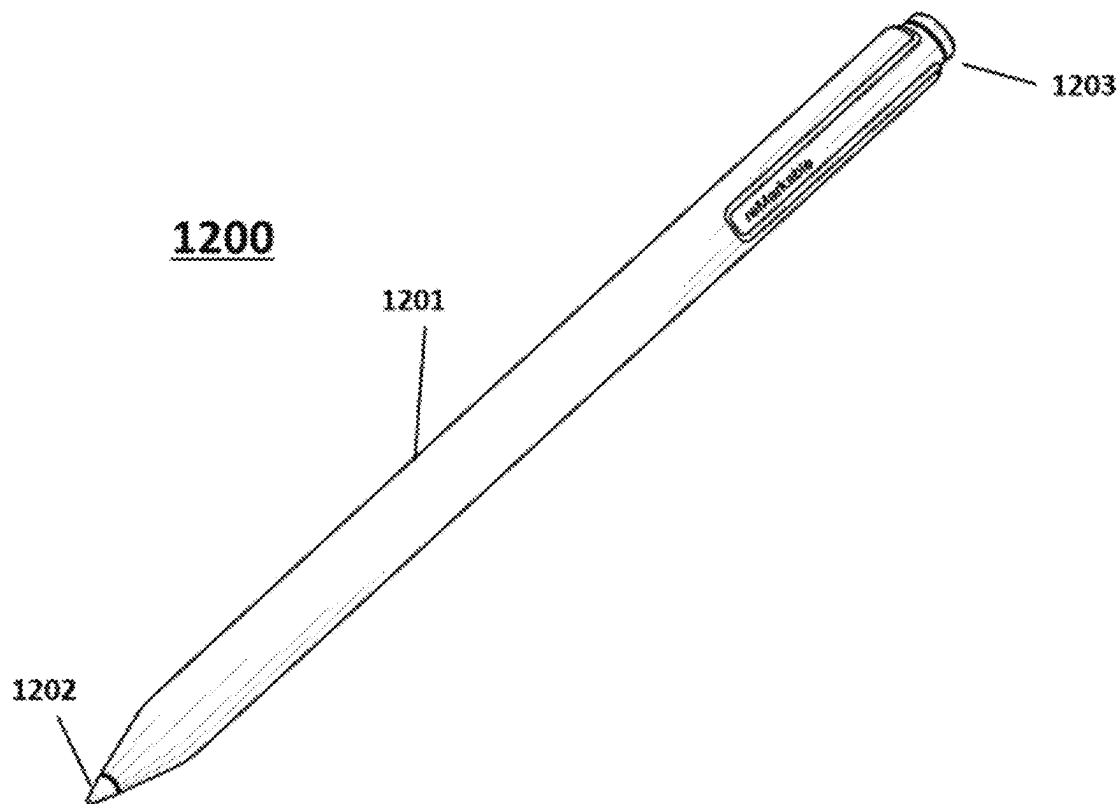


FIG. 12

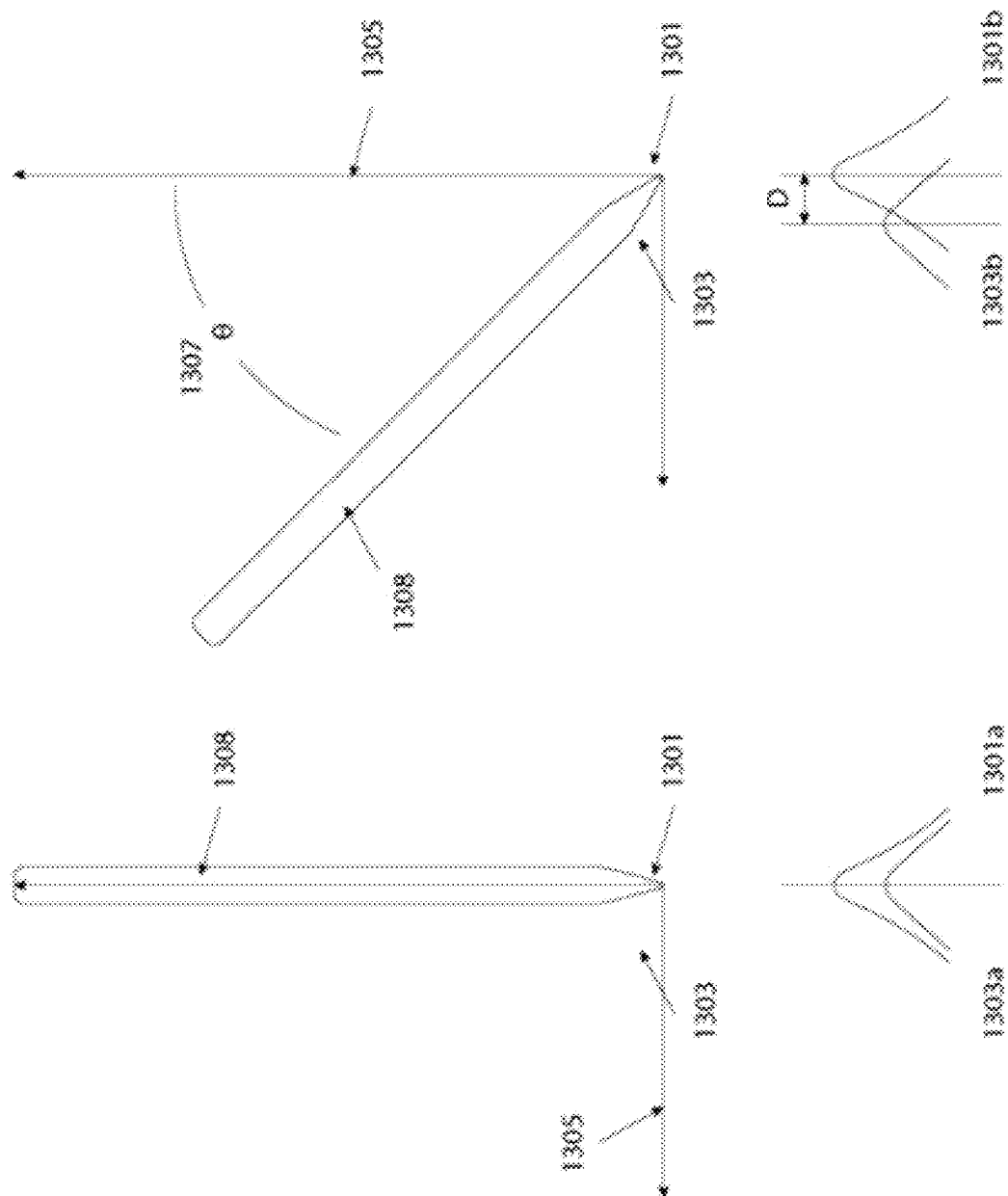


FIG. 13B

FIG. 13A

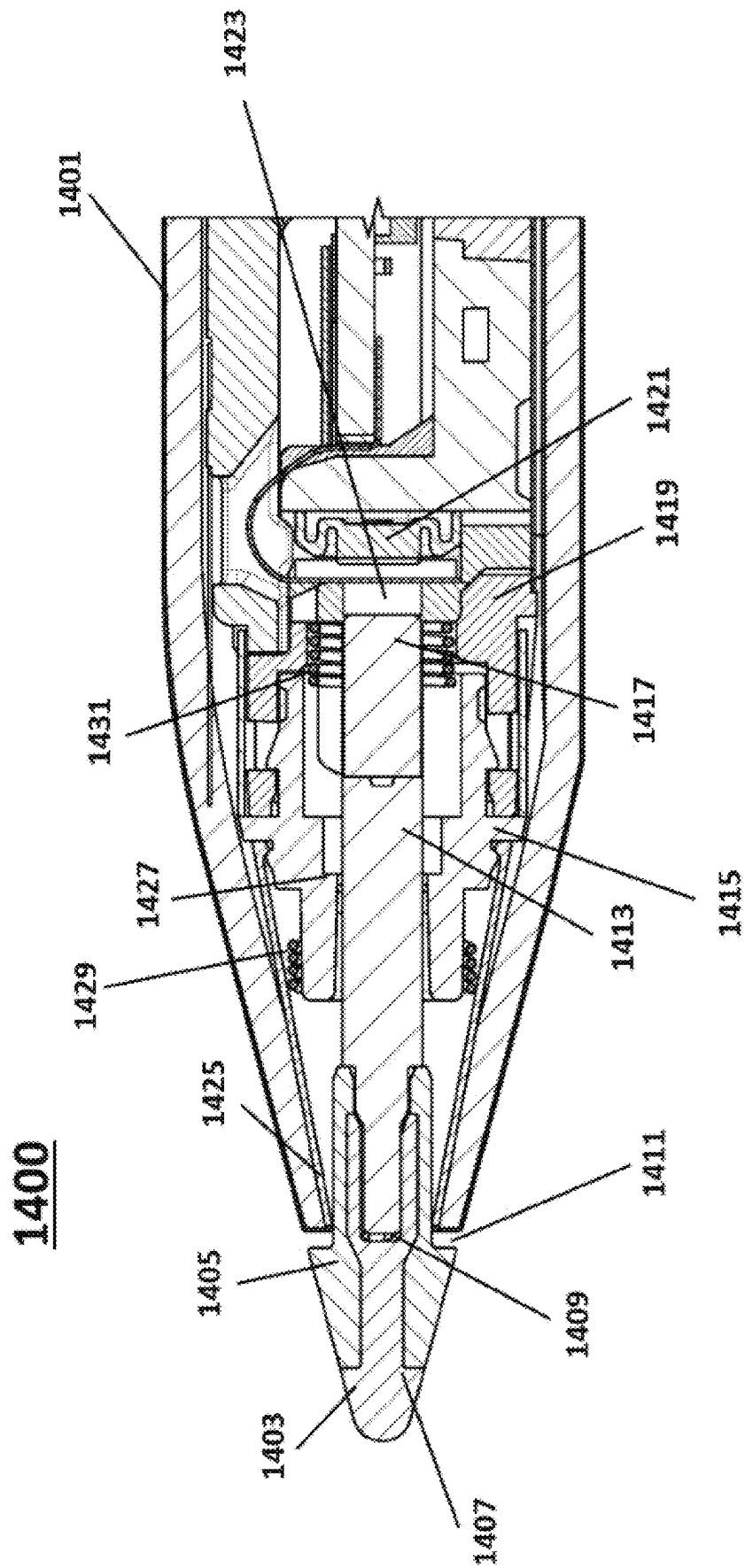


Fig. 14

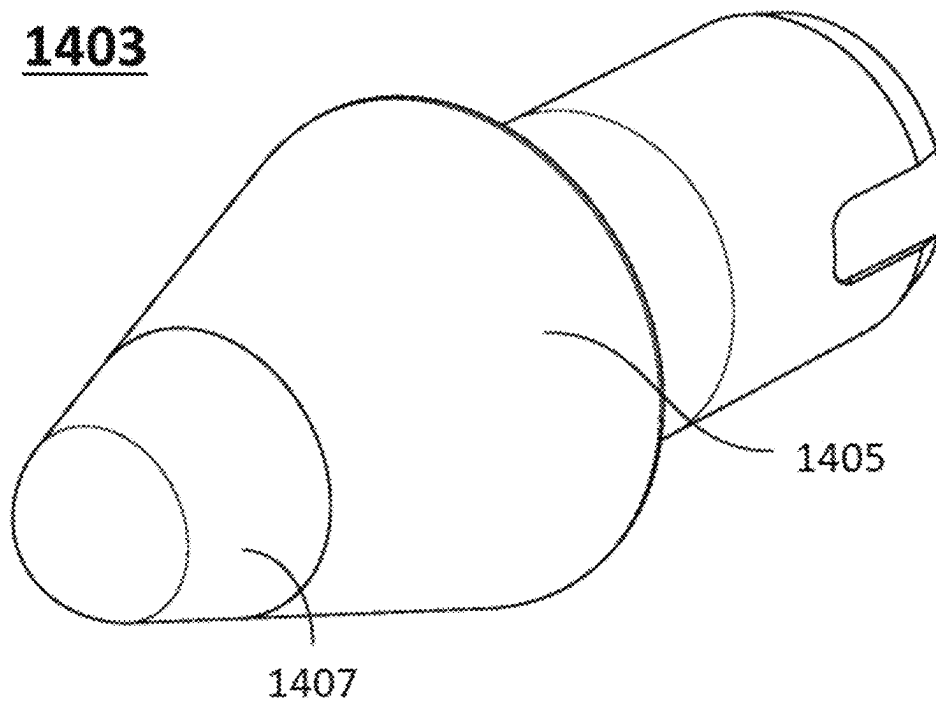


FIG. 15

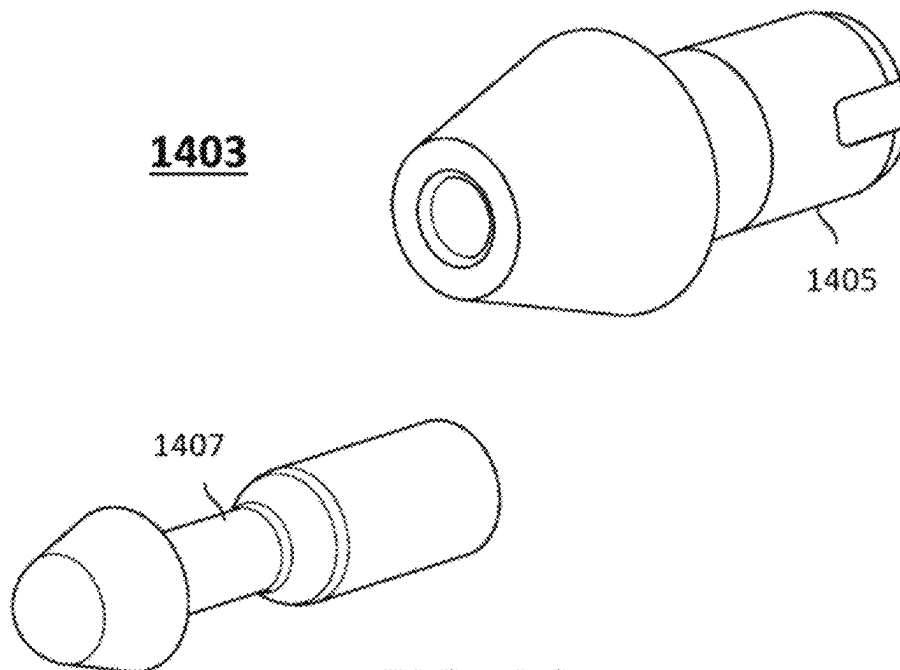


FIG. 16

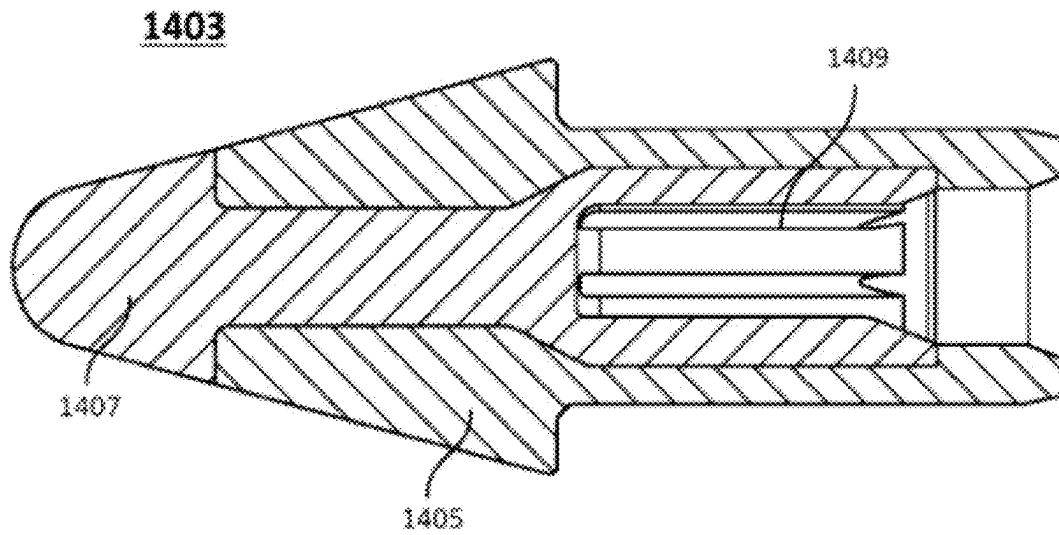


FIG. 17

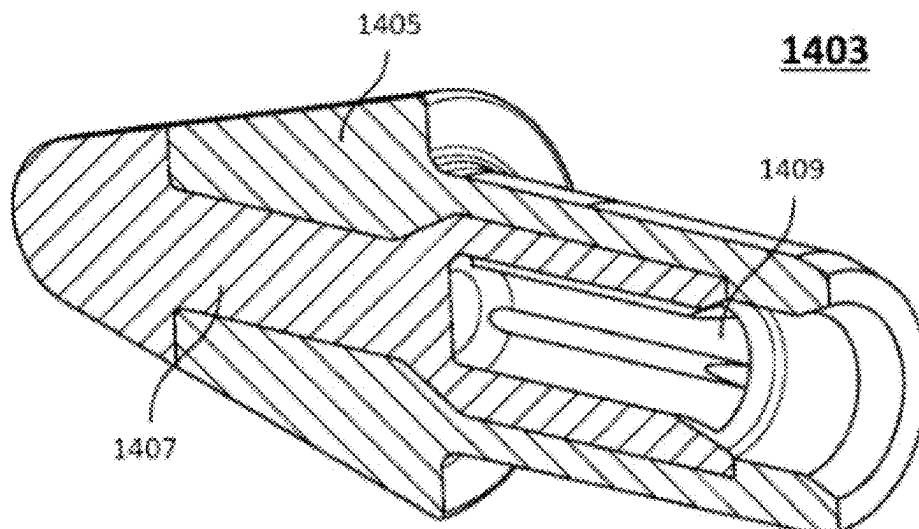


FIG. 18

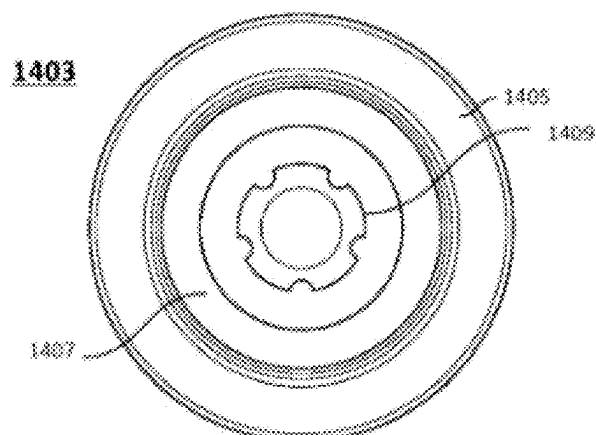


FIG. 19

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REPLACEABLE CONDUCTIVE MARKER TIP

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application, U.S. patent application Ser. No. 18/779,151, entitled "Replaceable Conductive Marker Tip" is related to U.S. patent application Ser. No. 18/779,154, entitled "Advanced Paper Emulation," filed Jul. 22, 2024; U.S. patent application Ser. No. 18/779,158 entitled "Marker Protection System," filed Jul. 22, 2024; U.S. patent application Ser. No. 18/779,164 entitled "Marker Writing System," filed Jul. 22, 2024; U.S. patent application Ser. No. 18/779,170 entitled "Captive Object Flexure Mechanism," filed Jul. 22, 2024; and U.S. patent application Ser. No. 18/779,149, entitled "Active Pen-Stylus Precise Eraser" filed Jul. 22, 2024, all of which are owned by the Applicant of the present application. These related applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

The disclosure relates generally to a pointing device, adapted for various coordinate input devices such as a digitizer or a tablet, which provide inputs to various types of computing systems. In particular, embodiments of the present invention relate to a replaceable marker tip that combines an antenna with an insulating material to provide a high quality writing feeling to users of the marker.

BACKGROUND

Mobile telephones, tablet computers, PCs, car entertainment systems, white goods and many other devices are commonly equipped with interactive displays. These interactive displays combine a display screen, such as an LCD, OLED, plasma or electrophoretic display (EPD), with an input system, such as a touch- or pen-stylus-input system. The input system recognizes the presence of an input object such as a pen-stylus touching or in proximity to the display screen. The device typically responds to such inputs by performing one or more functions, which may include changing what is shown on the display screen.

A "pen-stylus" (or "pen" or "stylus") is typically a pen- or pencil-shaped instrument whose position (e.g., tip position) on a computer monitor can be detected either electronically or physically. The pen-stylus enables users to perform tasks, such as drawing or making selections on a computing device. While devices with touchscreens, such as some computers, mobile devices (smartphones and personal digital assistants), game consoles, and graphics tablets, can often be operated with a fingertip, a pen-stylus typically provides more accurate and controllable input. In essence, a pen-stylus has a similar function as a mouse or touchpad as a pointing device but may enable much more precise inputs for certain drawing tasks. The use of a pen-stylus is sometimes termed "pen-stylus computing."

Conventional pen-styluses have typically been constructed to detect "pen-down" information in addition to coordinate information on the pointing device. Such pen-down information typically arises when the pen-stylus point is in contact with a panel of the digitizer. The pen-down information is conventionally detected by either force (e.g., pressure) sensitive means for detecting the vertical force applied to the pen-stylus point and/or detected by an electrical connection between the pen-stylus and the panel of the

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digitizer. The position data may be smoothed and/or de-noised before it is used to estimate the velocity and/or the acceleration of the input object. Such smoothing and/or de-noising may be done using an appropriate technique—for example, by applying a recursive Bayesian filter or smoothing, such as a Kalman filter, to the position data.

Active pen-styluses (also known as "active pen" or "digital styluses") include digital components and/or circuitry inside the pen-stylus that communicates with a digitizer on the touch device. This communication allows for advanced features such as force (e.g., pressure) sensitivity, tilt detection, programmable buttons, palm detection; eraser tips, memorizing settings, and writing data transmission.

Active pen-styluses typically employ different protocols from different manufacturers in order to communicate with the digitizer of a graphic tablet or multi-touch device. For an active pen-stylus to function properly, its digital component protocol must typically match the digitizer technology in the touch screen with which it interacts. Thus, the digital protocol of the pen-stylus must be compatible with the device digitizer, otherwise input from the pen-stylus will not register on the device. Active pen-styluses are typically powered by a removable or chargeable battery.

A pen-stylus' performance is often measured by four characteristics: 1) comfort, 2) resistance, 3) balance and overall weight, and 4) precision. "Precision" can sometimes be a nebulous characteristic, so it is often described in terms of further characteristics, such as: 1) responsiveness and speed, 2) jitter, 3) tilt, 4) levels of force (e.g., pressure), and 5) palm rejection or detection. This last element of precision may prevent a touch device from registering or marking the screen when a hand or palm is resting on the screen surface. Effective operation may rely on a combination of technology in the pen-stylus, the operating system software and the screen digitizer technology for effective operation.

While pen-stylus technology has made great strides in recent years in improving pen-stylus technology, further improvements are still warranted. Moreover, specific use cases for pen-styluses may compel levels of precision and additional functionality not available in conventional devices.

SUMMARY OF THE INVENTION

Embodiments of the invention provide a removable marker tip for an active pen-stylus, the removable marker tip having an antenna that sends signals originating in the active pen-stylus to a tablet device and an insulator, wherein the antenna and the insulator comprise consumable materials that enable the active pen-stylus to emulate for users sensations similar to applying pen or pencil to paper when using the active pen-stylus to draw on a display of the tablet device, wherein the removable marker tip transmits the physical forces it receives when the removable marker tip engages with the display on the tablet device to the active pen-stylus, wherein a force sensor translates the physical force received into an electronic signal that is transmitted back to the tablet device.

The antenna in the removable marker tip has a substantially spherical external shape, according to embodiments of the invention. The removable marker tip may have also crush ribs fitted into a posterior opening on the removable marker tip, the crush ribs fitting onto a writing shaft of the active pen-stylus and securing the removable marker tip to the writing shaft. In addition, the removable marker tip's antenna may comprise a conductive carbon fiber material.

Embodiments of the invention also provide an active pen-stylus having a removable marker tip that comprises a first antenna and an insulator. A writing shaft is adapted to receive and hold the removable marker tip to the active pen-stylus. A force sensor receives physical forces imparted to the writing shaft by the removable marker tip when the removable marker tip engages with a display on a tablet device and translates the physical force received into an electronic signal, wherein the electronic signal is transmitted back to the tablet device.

BRIEF DESCRIPTION OF DRAWINGS

The disclosed embodiments have other advantages and features which will be more readily apparent from the detailed description, the appended claims, and the accompanying figures (or drawings). A brief introduction of the figures is below.

FIG. 1 illustrates a system architecture for an e-paper tablet device 110 that receives inputs from the input mechanism such as a pen-stylus, according to one example embodiment.

FIG. 2 is a block diagram of the system architecture of an e-paper tablet device 110, according to one example embodiment.

FIG. 3 illustrates a front and right perspective view of an e-paper tablet 300 having the functionality described for the e-paper tablet device 110 in FIGS. 1-2.

FIG. 4 illustrates hardware components of an example Electrophoretic Display (EPD) operating in an e-paper tablet device 110, according to one example embodiment.

FIG. 5 is a block diagram illustrating components of an example machine able to read instructions from a machine-readable medium and execute them in a processor (or controller), according to one example embodiment.

FIG. 6 illustrates a rear view of the e-paper tablet 300 showing volcano feet 601a-601d, a pogo pad 603, and an antenna region 605, according to an embodiment of the invention.

FIG. 7 illustrates a top view of the e-paper tablet device 300 showing volcano feet 601a, 601d, and a power button 701, according to an embodiment of the invention.

FIG. 8 illustrates a bottom view of the e-paper tablet device 300 showing volcano feet 601b, 601d and the USB-c connector 307, according to an embodiment of the invention.

FIG. 9 illustrates a right view of the e-paper tablet device 300 showing volcano feet 601b, 601a, and the charging area 304 for recharging the input device 120, when the input device is an active pen-stylus, according to an embodiment of the invention.

FIG. 10 illustrates a left view of the e-paper tablet device 300 showing volcano feet 601c, 601d, according to an embodiment of the invention.

FIG. 11 illustrates a pen-stylus 1100 suited for application in at least one embodiment of the invention.

FIG. 12 illustrates an external casing 1201 for a pen-stylus 1200 designed to fit a user's hand, according to an embodiment of the invention.

FIGS. 13A-13B illustrate two transmitters 1301, 1303 in a marker's tip (e.g., the core 1102 shown in FIG. 11) of an active pen-stylus 1308 giving a computing device (e.g., the e-paper tablet 110) the ability to measure two distinct signals.

FIG. 14 illustrates a cross section of the forward portion of an active pen-stylus 1400 that a user may employ for tasks

such as drawing a line on the display of a device (e.g., the e-paper tablet 110), according to an embodiment of the invention.

FIG. 15 illustrates a replaceable marker tip 1403 for the pen-stylus 1400 shown in FIG. 14 that provides a good writing feeling and a high quality antenna function, according to an embodiment of the invention.

FIG. 16 illustrates a marker tip 1403 comprising a core antenna 1407 and an insulating material 1405, according to an embodiment of the invention.

FIGS. 17-19 illustrate crush ribs 1409 that hold the core antenna 1407 onto the writing shaft 1413, according to an embodiment of the invention.

The figures depict various embodiments of the presented invention for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles described herein.

DETAILED DESCRIPTION

The Figures (FIGS.) and the following description relate to preferred embodiments by way of illustration only. It should be noted that from the following discussion, alternative embodiments of the structures and methods disclosed herein will be readily recognized as viable alternatives that may be employed without departing from the principles of what is claimed.

Reference will now be made in detail to several embodiments, examples of which are illustrated in the accompanying figures. It is noted that wherever practicable similar or like reference numbers may be used in the figures and may indicate similar or like functionality. The figures depict embodiments of the disclosed system (or method) for purposes of illustration only. An ordinarily skilled artisan will readily recognize from the following description that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles described herein.

Overview

Disclosed is a system and related process) for a replaceable marker tip in an active pen-stylus for application with a tablet device, particularly an e-paper tablet device. The replaceable marker tip is intended to provide a good (paper-like) writing feeling for users of the e-paper tablet device while also providing a high quality antenna function. A description of this specific invention begins at FIG. 15, following a description of a related pen-stylus at FIG. 11. Before describing the invention in greater detail, a description will be provided of the e-paper tablet with which embodiments of the pen-stylus interact, as well as a description of an embodiment of the pen-stylus itself. Following this description, embodiments of the marker tip invention will be described.

Example System and Device Configuration—for Supporting Tablet Device

As shown in FIG. 1, an e-paper tablet device 110 receives inputs from the input mechanism 120, for example, when the input mechanism 120 makes physical contact with a contact-sensitive surface (e.g., the touch-sensitive screen) on the e-paper tablet device 110 as the user makes a gesture of some sort with the input mechanism 120. The input mechanism 120 may be a finger, pen-stylus or marker. The tablet device 110 here is referred to as an “e-paper tablet,” a device that mimics the feeling of writing with ordinary pen and paper for users of the device. Such devices are also known

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as “electronic paper” and “electronic ink”. Based on the nature of the contact, the e-paper tablet device **110** generates and executes instructions for updating content displayed on the contact-sensitive screen to reflect the gesture inputs. For example, in response to a gesture transcribing a verbal message (e.g., a written text or a drawing), the e-paper tablet device **110** updates the contact-sensitive screen to display the transcribed message. As another example, in response to a gesture selecting a navigation option, the e-paper tablet device **110** updates the screen to display a new page associated with the navigation option. While embodiments of the invention have been designed for e-paper systems, embodiments of the invention may also be suitable for other forms of computing devices capable of receiving and processing inputs from pen-stylus devices.

The input mechanism **120** may refer to any device or object that is compatible with the contact-sensitive screen of the e-paper tablet device **110**, in particular a pen-stylus device, such as a so-called active pen device having its own power source or a static pen that receives its power from engagement with the contact-sensitive screen on the e-paper tablet device **110**. In one embodiment, the input mechanism **120** may work with an electronic ink (e.g., E-ink) contact-sensitive screen. For example, the input mechanism **120** may refer to any device or object that can interface with a screen and, from which, the screen can detect a touch or contact of said input mechanism **120**. Once the touch or contact is detected, electronics associated with the screen generate a signal which the e-paper tablet device **110** can process as a gesture that may be provided for display on the screen. Upon detecting a gesture by the input mechanism **120**, electronics within the contact-sensitive screen generate a signal that encodes instructions for displaying content or updating content previously displayed on the screen of the e-paper tablet device **110** based on the movement of the detected gesture across the screen. For example, when processed by the e-paper tablet device **110**, the encoded signal may cause a representation of the detected gesture to be displayed on the screen of the e-paper tablet device **110**, such as a scribble. As mentioned, the input mechanism **120** may be a pen-stylus or another type of pointing device, including a part of a user’s body, such as a finger.

In one embodiment, the input mechanism **120** is an encased magnetic coil. When in proximity to the screen of the e-paper tablet device **110**, the magnetic coil helps generate a magnetic field that encodes a signal that communicates instructions, which are processed by the e-paper tablet device **110** to provide a representation of the gesture for display on the screen, e.g., as a marking. The input mechanism **120** may be force (e.g., pressure) and tilt-sensitive such that the system can make natural, visual response to both the pressure and tilt applied by the user. In turn, the interaction between the input mechanism and the contact-sensitive screen of the e-paper tablet device **110** may generate a different encoded signal for processing, for example, to provide for display a representation of the gesture on the screen that has different characteristics, e.g., thicker line marking. In alternate embodiments, the input mechanism **120** includes a power source (e.g., a battery) which can generate an electric field with a contact-sensitive surface. It is noted that the encoded signal is a signal that is generated and may be communicated. The encoded signal may have a signal pattern that may be used for further analog or digital analysis (or interpretation).

In one embodiment, the contact-sensitive screen is a capacitive touchscreen. The screen may be designed using a glass or polymer material coated with a conductive material.

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Electrodes, or an alternate current carrying electric component, are arranged along the coating of the screen (e.g., in a diamond-shaped cross hatch) to maintain a constant level of current running throughout the screen. A second set of electrodes are arranged horizontally. The matrix of vertical active electrodes and horizontal inactive electrodes generates an electrostatic field at each point on the screen. When an input mechanism **120** with conductive properties, for example the encased magnetic coil, a human finger, or something else that triggers the capacitive effect, is brought into contact with an area of the screen of the e-paper tablet device **110**, current flows through the horizontally arranged electrodes, disrupting the electrostatic field at the contacted point on the screen. The disruption in the electrostatic field at each point that a gesture covers may be measured, for example as a change in capacitance, and encoded into an analog or digital signal.

In an alternate embodiment, the contact-sensitive screen is a resistive touchscreen. The resistive touch screen comprises two metallic layers: a first metallic layer in which striped electrodes are positioned on a substrate, such as a glass or plastic and a second metallic layer in which transparent electrodes are positioned. When contact from an input mechanism, for example a pen-stylus, finger, or palm, is made on the surface of the touchscreen, the two layers are pressed together. Upon contact, a voltage gradient is applied to the first layer and measured as a distance by the second layer to determine a horizontal coordinate of the contact on the screen. The voltage gradient is subsequently applied to the second layer to determine a vertical coordinate of the contact on the screen. The combination of the horizontal coordinate and the vertical coordinate register an exact location of the contact on the contact-sensitive screen. Unlike capacitive touchscreens which rely on conductive input mechanisms, a resistive touchscreen is configured to sense contact from nearly any input mechanism. Although some embodiments of the e-paper tablet device **110** are described herein with reference to a capacitive touchscreen, one skilled in the art would recognize that a resistive touchscreen could also be implemented.

In an alternate embodiment, the contact-sensitive screen is an inductive touchscreen. An inductive touchscreen comprises a metal front layer that is configured to detect deflections when contact is made on the screen by an input mechanism. Accordingly, an inductive touchscreen is configured to sense contact from nearly any input mechanism. Although some embodiments of the e-paper tablet device **110** are described herein with reference to a capacitive touchscreen, an ordinarily skilled artisan would recognize that alternative touchscreen technology may be implemented, for example, an inductive touchscreen could also be implemented.

The cloud server **130** is configured to receive information from the e-paper tablet device **110** and/or communicate instructions to the e-paper tablet device **110**, according to some embodiments of the invention. As illustrated in FIG. 1, the cloud server **130** may comprise a cloud data processor **150** and a data store **160**. Data recorded and stored by the e-paper tablet device **110** may be communicated via the network **140** to the cloud server **130** for storage in the data store **160**. For example, the data store **160** may store documents, images, or other types of content generated or recorded by a user through the e-paper tablet device **110**. In some embodiments, the cloud data processor **150** monitors the activity and usage of the e-paper tablet device **110** and communicates processing instructions to the e-paper tablet device **110**. For example, the cloud data processor **150** may

regulate synchronization protocols for data stored in the data store **160** with the e-paper tablet device **110**.

Interactions between the e-paper tablet device **110** and the cloud server **130** are typically performed via the network **140**, which enables communication between the e-paper tablet device **110** and the cloud server **130**. In one embodiment, the network **140** uses standard communication technologies and/or protocols including, but not limited to, links using technologies such as Ethernet, 802.11, worldwide interoperability for microwave access (WiMAX), 3G, 4G, LTE, digital subscriber line (DSL), asynchronous transfer mode (ATM), InfiniBand, and PCI Express Advanced Switching. The network **140** may also utilize dedicated, custom, or private communication links. The network **140** may comprise any combination of local area and/or wide area networks, using both wired and wireless communication systems. The cloud server **130** may be alternatively implemented, and in some embodiments may be replaced by hardware and software that provide similar functionality while possibly not being considered a conventional cloud server.

FIG. 2 is a block diagram of the system architecture of an e-paper tablet device **110**, according to one example embodiment. In the embodiment illustrated in FIG. 2, the e-paper tablet device **110** comprises an input detector module **210**, an input digitizer **220**, a display system **230**, and a graphics generator **240**.

The input detector module **210** recognizes that a gesture has been or is being made on the screen of the e-paper tablet device **110**. The input detector module **210** refers to electronics integrated into the screen of the e-paper tablet device **110** that are configured to interpret an encoded signal generated by contact between the input mechanism **120** and the screen into a recognizable gesture. To do so, the input detector module **210** may evaluate properties of the encoded signal to determine whether the signal represents a gesture made intentionally by a user or a gesture made unintentionally by a user.

The input digitizer **220** may be configured to convert the analog signal encoded by the contact between the input mechanism **120** and the screen into a digital set of instructions. The converted digital set of instructions may be processed by the e-paper tablet device **110** to generate or update a user interface displayed on the screen to reflect an intentional gesture.

The display system **230** may include the physical and firmware (or software) components to provide for display (e.g., render) on a screen a user interface. The user interface may correspond to any type of visual representation that may be presented to or viewed by a user of the e-paper tablet device **110**.

Based on the digital signal generated by the input digitizer **220**, the graphics generator **240** may be configured to generate or update graphics of a user interface to be displayed on the screen of the e-paper tablet device **110**. The display system **230** may be configured to present those graphics of the user interface for display to a user using electronics integrated into the screen.

When an input mechanism **120** makes contact with a contact-sensitive screen of an e-paper tablet device **110**, the input detector module **210** recognizes a gesture has been made through the screen. The gesture may be recognized as a part of an encoded signal generated by a pressure or force sensor in the input mechanism **120** and/or corresponding electronics of the screen of the display system **230**. The encoded signal is transmitted to the input detector module **210**, which evaluates properties of the encoded signal in

view of at least one gesture rule to determine whether the gesture was made intentionally by a user. If the input detector module **210** determines that the gesture was made intentionally, the input detector module **210** communicates the encoded signal to the digitizer output. The encoded signal is an analog representation of the gesture received by a matrix of sensors embedded in the screen of the device **110**.

In one example embodiment, the input digitizer **220** translates the physical points on the screen that the input mechanism **120** made contact with into a set of instructions for updating what is provided for display on the screen. For example, if the input detector module **210** detects an intentional gesture that swipes from a first page to a second page, the input digitizer **220** receives the analog signal generated by the input mechanism **120** as it performs the swiping gesture. The input digitizer **220** generates a digital signal for the swiping gesture that provides instructions for the display system **230** of the e-paper tablet device **110** to update the user interface of the screen to transition from, for example, a current (or first page) to a next (or second page, which may be before or after the first page).

In one example embodiment, the graphics generator **240** receives the digital instructional signal, such as a swipe gesture indicating page transition (e.g., flipping or turning) generated by the input digitizer **220**. The graphics generator **240** generates graphics or an update to the previously displayed user interface graphics based on the received signal. The generated or updated graphics of the user interface are provided for display on the screen of the e-paper tablet device **110** by the display system **230**, e.g., displaying a transition from a current page to a next page to a user. In the displayed embodiment of the FIG. 2, the graphics generator **240** comprises a rasterizer module **250** and a depixelator module **260**. Input gestures drawn by a user on a contact-sensitive surface are received as vector graphics and are input to the rasterizer module **250**. The rasterizer module **250** converts the input vector graphics to raster graphics, which can be displayed (or provided for display) on the contact-sensitive surface. The depixelator module **260** may apply image processing techniques to convert the displayed raster graphics back into vector graphics, for example to improve processing power of the e-paper tablet device **110** and to conserve memory of the e-paper tablet device **110**. In at least one implementation, the depixelator module **260** may convert a displayed raster graphic back to a vector graphic when exporting content displayed on the screen into a different format or to a different system.

Further details about structures and functions of e-paper tablets and their graphical displays can be found in U.S. Pat. No. 11,158,097 to Martin Sandsmark and Gunnar Sletta entitled "Generating vector graphics by processing raster graphics" and in U.S. Pat. No. 10,824,274 to Sondre Hoff Dyvik, Martin Sandsmark, and Magnus Haug Wanberg, entitled "Interactive displays," both of which are incorporated by reference herein.

FIG. 3 illustrates a front and right perspective view of an e-paper tablet **300** having the functionality described for the e-paper tablet device **110** in FIGS. 1-2. Among other things, the e-paper tablet **300** includes a touch-sensitive display **303**. The display **303** has been treated to provide a paper-feeling for users of the device when they engage with it using an input device **120**. FIG. 3 also shows a charging area **304** for recharging the input device **120**, when the input device is an active pen-stylus, according to an embodiment of the invention. Inside the e-paper tablet **300** near where the charging area **304** is located may be a set of magnets to hold the input

device **120** in place while it is re-charging. FIG. **3** also shows a USB-c connector **307** that may be used to provide electrical power to the e-paper tablet **300**, as well as transmitting various types of data into or out of the e-paper tablet **300**. The e-paper tablet **300** also includes several actuators and other features that will be shown below in FIGS. **5-10**.

FIG. **4** illustrates hardware components of an example Electrophoretic Display (EPD) in accordance with a disclosed embodiment. As discussed, a variety of display technologies may be employed, including EPDs, LCDs, and reflective LCDs (rLCDs). The specific display device deployed may be part of the display system **230** of the e-paper tablet device **110** shown in FIG. **2** and produce the images shown on the display **303** of the e-paper tablet **300** shown in FIG. **3**. The EPD includes a gate driver **409**, a source driver **411**, a shift register **423** with data and clock signal line, a latch **425**, a voltage selector **427**, and rows making up a display **405**. The EPD industry borrowed certain components and concepts from the LCD industry; however, these two devices have some fundamental differences as well. Of particular relevance here is the persistence of pixels in EPD displays. Unlike LCD displays, EPD displays do not require the frequent refreshing required in an LCD display. In an EPD display, once a neutral voltage is set for a pixel, the pixel will not change, for example, and will persist for a long period of time, especially relative to an LCD display.

As mentioned, Electrophoretic displays (EPDs) **405** have utilized many aspects of LCD production infrastructure and driving mechanisms. The driving electronics typically consist of a gate driver (GD) **409** and a source driver (SD) **411**. The display **405** has multiple rows of pixels. Pixel values within a row may be changed, e.g., logic high voltage may be a “black” pixel and a logic low voltage or “ground” may be a no color pixel. The pixels in the EPD **405** function similarly to small capacitors that persist over long time intervals. An EPD pixel contains a large number of charged particles that are suspended in a liquid. If a charge is applied, the particles will move to a surface where they become visible. White and black particles have opposite charges such that a pixel’s display may change from white to black by applying an opposite charge to the pixel. Thus, the waveforms applied to an EPD comprise long trains of voltages to change from black to white or vice versa. The EPD arts are also known to have the ability to apply variable voltage levels that mix the white and black particles to produce various shades of gray. Voltage levels in a pixel also may be tiered between to provide shades between no color and black (e.g., levels of grey). Groups of pixels around each other may form a region that provides some visible characteristic to a user, e.g., an image on a screen, e.g., of the display system **230** of the e-paper tablet device **110**.

To change pixel values in a region, a scan of a display **405** will conventionally start at a top row, e.g., row **0** **421**, and apply voltages to update pixels within a particular row where pixels need to be changed to correspond with the image that is displayed. In this example, a start pulse (GDSP) **403** can be used to reset the driver **411** to row **0** **421**, and a direction (DIR) **404** can be used to reset a direction. A row-by-row selection is made by driving the driver gate **409** to select a row, e.g., active row **413**. All pixels in one row are addressed concurrently using data transferred to the display. Latch **425** receives from the shift register **423** the next set of voltages to be applied to a row of pixels. When the scan of the active row is completed and, if necessary,

pixels changed or updated, a clock pulse (GDCLK) **415** is issued to the driver gate **409** to change to the next row **417** for a scan.

As mentioned above, an ordinary artisan will recognize that a similar function can be accomplished also with a standard LCD, OLED, MicroLED or other type of display, and the description of EPD technology is provided here merely for illustration of one embodiment of the invention.

The source driver **411** is used to set the target voltage for each of the pixels/columns for the selected row. It consists of a shift register **423** for holding the voltage data, a latch circuit **425** for enabling pixel data transfer while the previous row is being exposed, and a voltage selector (multiplexer) **427** for converting the latched voltage selection into an actual voltage. For all rows to be updated all the voltage values have to be shifted into the register **423** and latched for the voltages to be available.

FIG. **5** is a block diagram illustrating components of an example machine able to read instructions from a machine-readable medium and execute them in a processor (or controller), according to one embodiment. In this example, FIG. **5** shows a diagrammatic representation of a machine in the example form of a computer system **500** (e.g., the computing portions of the e-paper tablet **111** shown in FIG. **1**) within which program code (e.g., software) for causing the machine to perform any one or more of the methodologies discussed herein may be executed. The e-paper tablet device **110** may include some or all of the components of the computer system **500**. The program code may be comprised of instructions **524** executable by one or more processors **502**. In the e-paper tablet system **110**, the instructions may correspond to the functional components described in FIGS. **1**, **2**, and **4**. FIG. **5** is an example of a processing system, of which a some of the described components or all of the described components may be leveraged by the modules described herein for execution.

While the embodiments described herein are in the context of the e-paper tablet system **110**, it is noted that the principles may apply to other touch sensitive devices. In those contexts, the machine of FIG. **5** may be a server computer, a client computer, a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a cellular telephone, a smartphone, a web appliance, a network router, an internet of things (IoT) device, a switch or bridge, or any machine capable of executing instructions **524** (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute instructions **524** to perform any one or more of the methodologies discussed herein.

The example computer system **500** includes one or more processors **502** (e.g., a central processing unit (CPU), one or more graphics processing units (GPU), one or more digital signal processors (DSP), one or more application specific integrated circuits (ASICs), one or more radio-frequency integrated circuits (RFICs), or any combination of these), a main memory **504**, and a static memory **506**, which are configured to communicate with each other via a bus **508**. The computer system **500** may further include visual display interface **510**. The visual interface may include a software driver that enables displaying user interfaces on a screen (or display). The visual interface may display user interfaces directly (e.g., on the screen) or indirectly on a surface, window, or the like (e.g., via a visual projection unit). For ease of discussion the visual interface may be described as a screen or display screen. The visual interface **510** may

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include or may interface with a touch enabled screen, e.g., of the e-paper tablet system **110** and may be associated with the display system **230**. The computer system **500** may also include an input device **512** (e.g., a pen-stylus, a keyboard, or touch screen keyboard), a cursor control device **514** (e.g., a mouse, a trackball, a joystick, a motion sensor, or other pointing instrument), a storage unit **516**, a signal generation device **518** (e.g., a speaker), and a network interface device **520**, which also are configured to communicate via the bus **508**.

The storage unit **516** includes a machine-readable medium **522** on which is stored (or encoded) instructions **524** (e.g., software) embodying any one or more of the methodologies or functions described herein. The instructions **524** (e.g., software) may also reside, completely or at least partially, within the main memory **504** or within the processor **502** (e.g., within a processor's cache memory) during execution thereof by the computer system **500**, the main memory **504** and the processor **502** also constituting machine-readable media. The instructions **524** (e.g., software) may be transmitted or received over a network **426** via the network interface device **520**.

While machine-readable medium **522** is shown in an example embodiment to be a single medium, the term "machine-readable medium" should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store instructions (e.g., instructions **524**). The term "machine-readable medium" shall also be taken to include any medium that is capable of storing instructions (e.g., instructions **524**) for execution by the machine and that cause the machine to perform any one or more of the methodologies disclosed herein. The term "machine-readable medium" includes, but not be limited to, data repositories in the form of solid-state memories, optical media, and magnetic media.

The computer system **500** also may include the one or more sensors **525**. Also note that a computing device may include only a subset of the components illustrated and described with FIG. **5**. For example, an IoT device may only include a processor **502**, a small storage unit **516**, a main memory **504**, a visual interface **510**, a network interface device **520**, and a sensor **525**.

Representative E-Paper Tablet

FIG. **3** provided a representative view of an e-paper tablet **300**, resembling the e-paper **110** shown in FIGS. **1-2**. FIG. **6** illustrates a rear view of the e-paper tablet **300** showing volcano feet **601a-601d**, a pogo pad **603**, and an antenna region **605**, according to an embodiment of the invention. The antenna region **605** resides outside and above the location for a main antenna (e.g., an antenna on the e-paper tablet **110** that communicates with the cloud server **130** shown in FIG. **1** and which may generate the e-paper tablet's beacon signal discussed below with respect to FIG. **14**) for the e-paper tablet **300**, allowing the e-paper tablet device **300** to connect to the Internet, for example. The pogo pad **603** allows the e-paper tablet device **300** to connect to other devices, such as a folio device having a keyboard, for example.

FIG. **7** illustrates a top view of the e-paper tablet device **300** showing volcano feet **601a**, **601d**, and a power button **701**, according to an embodiment of the invention.

FIG. **8** illustrates a bottom view of the e-paper tablet device **300** showing volcano feet **601b**, **601d** and the USB-c connector **307**, according to an embodiment of the invention.

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FIG. **9** illustrates a right view of the e-paper tablet device **300** showing volcano feet **601b**, **601a**, and the charging area **304** for recharging the input device **120**, when the input device is an active pen-stylus, according to an embodiment of the invention. Inside the e-paper tablet **300** near where the charging area **304** is located may be a set of magnets to hold the input device **120** in place while it is re-charging.

FIG. **10** illustrates a left view of the e-paper tablet device **300** showing volcano feet **601c**, **601d**, according to an embodiment of the invention.

Example Active Pen in Operation

An active pen-stylus (or more commonly "active pen") is a pen-stylus input device that allows users to e.g., write, sketch or draw on the display of the computing device, e.g., the e-paper tablet **110**. An active pen-stylus includes digital components and/or circuitry that communicate with the computing device, e.g., the e-paper tablet. This communication enables advanced features such as force (e.g., pressure) sensitivity, tilt detection, programmable buttons, palm detection, eraser tips, memorizing settings, and writing data transmission. Viewed more expansively, communications between the computing device and the active pen-stylus enables a wide mix of peripheral sensors to be placed in the active pen-stylus with the resulting data reported to the computing device, e.g., the e-paper tablet. Such sensors placed in the active pen-stylus may range from simple buttons to enhanced artificial intelligence features.

An active pen's electronic components typically include a power source that may enable the device's electronics to provide lower latency and greater fidelity than other pen types, e.g., a passive pen. Active pens provide a number of advantages over passive pens, including hover latency, e.g., an active pen may typically be activated by merely being in proximity to a display, e.g., the display associated with the e-paper tablet **110**.

Once the active pen touches or contacts a display screen of a device like an e-paper tablet, electronics associated with the display screen generates a signal which the e-paper tablet (e.g., the e-paper tablet **110**) can process as a gesture made by the user. Upon detecting the gesture by the pen-stylus, electronics within the contact-sensitive screen generates a signal that encodes instructions for displaying content or updating content previously displayed on the screen of the e-paper tablet device based on the movement of the detected gesture across the screen.

In contrast with an active pen, a passive pen typically has no internal power source. A passive pen remains in an inactive state until the pen touches a device screen (e.g., a tablet device screen) causing a signal to pass from the device through the passive stylus and back to the device. The electronics associated with a passive pen may be integrated into the pen-stylus device or even provided in a small cartridge that is placed inside a pen-shaped stylus cover designed to better suit human ergonomics than the small cartridge containing the electronics and other components.

FIG. **11** illustrates an active pen-stylus **1100** that comprises a core member **1102** that itself comprises one or more antennas configured for communications with a tablet device, such as the e-paper tablet **110**. The active pen-stylus **1100** may include one or more force sensing systems **1104** that detect force, e.g., the forces exerted on a display of an e-paper tablet **110** by a user. In addition, the active pen-stylus **1100**. In addition, this element may also help emulate human interaction with conventional writing utensils like pencil and paper, e.g., render "a pencil and paper stack feeling". The active pen-stylus **1100** also includes a power source, e.g., a battery **1106**. Among other things, the battery

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1106 allows the active pen-stylus 1100 to support a “hover” function that allows the active pen-stylus 1100 to enter a sleep state for battery charge preservation when it is not actively engaged with the display of the e-paper tablet 110 and to wake up from the sleep state when the core member 1102 detects a proximity to the display of the e-paper tablet 110. The active pen-stylus 1100 might not actually draw lines on the display of the e-paper tablet 110 until a tip of the active pen-stylus 1100 physically touches the display of the e-paper tablet 110, according to an embodiment of the invention.

An active pen 1100 conventionally comprises a PCBA 1105 which includes electronic components needed for driving the signal lines associated with the core member 1102. The PCBA 1105 could alternatively be provided as a flexible printed circuit (FPC). The PCBA 1105 may include an appropriate active pen PCBA or IC/ASIC/MCU that processes data received from the core member 1104 for sensing force or displacement pressure with high resolution sensing of the force during the interaction between the active pen 1100 and an object, e.g., the surface of the computing device, such as the e-paper tablet 110).

The active pen 1100 may serve as the input mechanism 120 described in FIG. 1 that provides data input to the e-paper tablet 110, which may result in a drawing appearing on the display of the e-paper tablet 110. The active pen 1100 may also include an erasure system that receives user instructions related to erasing portions of a display on the screen of the associated computing device, e.g., the e-paper tablet 110. In a sense, the erasure system operates identically as the active pen-stylus system but where one draws, the other erases.

In operation, the force sensing system 1104 receives physical forces imparted to a marker tip of the active pen 1100 (e.g., the marker tip 1202 shown in FIG. 12) when the marker tip engages with a display on a tablet device (e.g., the e-paper tablet 110) and translates the physical force received into an electronic signal that is transmitted to the PCBA 1105 whose electronic components may perform a variety of processes on the signals received. The PCBA 1105 may then transmit the signals back to the tablet device (e.g., the e-paper tablet 110) for further action (e.g., drawing a line).

In embodiments where a secondary antenna system in the active pen 1100 enables tail eraser functionality, the active pen 1100 also comprises a second antenna system 1103, which enables the active pen 1100 to transmit and receive signals with the computing device (e.g., the e-paper tablet 110) through the tail eraser portion of the active pen 1100 near a cap 1108. In this embodiment, the active pen 1100 also includes a second force (e.g., pressure) sensing system 1107 that controls the force (e.g., pressure) imparted to the display of the computing device from the erasure activity. Likewise, the PCBA 1105 may include an appropriate active pen PCBA or IC/ASIC/MCU that process data associated with erasure functionality received from the second antenna system 1103.

The active pen 1100 also typically includes an external casing 1101 as a pen-stylus holder, typically formed in a cylindrical shape and made of non-metal material such as a plastic that contains the internal electronics within the casing 1101. The top end of the casing 1101 may be provided with the cap 1108.

FIG. 12 illustrates an external casing 1201 for a pen-stylus 1200 designed to fit a user’s hand, according to an embodiment of the invention. The pen-stylus 1200 includes a marker tip 1202 and an eraser 1203. The marker tip 1202 operates as a part of the core member 1102 shown in FIG.

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11, and the eraser 1203 operates as part of the second antenna system 1103 shown in FIG. 11.

When an active pen-stylus (e.g., the pen-stylus 1100 and/or the pen-stylus 1200) includes multiple antennas as part of the core (e.g., the core 1102), then the pen-stylus may provide additional capabilities. As shown in FIGS. 13A-13B, employing two separate transmitters 1301, 1303 in the marker tip (e.g., the core 1102 shown in FIG. 11) of the active pen-stylus 1308 gives the computing device (e.g., the e-paper tablet 110) the ability to measure two distinct signals (illustrated in FIG. 13A as 1301a, 1303a and illustrated in FIG. 13B as 1301b, 1303b). By knowing the antenna separation in the input device (e.g., the active pen-stylus 1308), the input digitizer 220 in a larger computing device, such as the e-paper tablet 110, can derive the active pen-stylus tilt angle θ 1307 relative to the computing device (e.g., the e-paper tablet 110). The input digitizer 220 is conventionally able to process active pen-stylus tilt information related to the drawing portion of the active pen-stylus. Only a slight change needs to be made to the input digitizer 220 for it to process tilt angle data related to an eraser (if erasure functionality is provided) to cause a change to the display on the e-paper tablet 110.

As seen in FIG. 13A, the signals 1301a, 1303a from the two antennas 1301, 1303 coincide when the active pen-stylus 1308 is perpendicular to the graphics display of the computing device (as shown by the grid 1305 and the line passing through the centers of both signals 1301a, 1303a). In contrast, FIG. 13B illustrates the active pen-stylus 1308 tilted by an angle θ 1307 relative to the display of the computing device (e.g., the e-paper tablet 110) as shown by the grid 1305 and the distance D marking the distance between the two centers of signals 1301b, 1303b. With such a tilt, the signals 1301b, 1303b from the two antennas 1301, 1303 are not aligned and are separated by the distance D. This distance D may be used to determine the corresponding drawing amount performed by the e-paper tablet device 110 on the display.

The method for deriving the tilt angle θ 1307 of the active pen-stylus 1308 is known when using two antennas (e.g., two separate antennas in the antenna system 1102 shown in FIG. 11, presented in FIGS. 13A and 13B as antennas 1301, 1303). The distance between antenna 1301 and antenna 1303 is fixed and known. This known separation, combined with basic trigonometry, can be employed in active pen tip antenna systems to derive the tilt angle for the active pen-stylus tip (e.g., the angle of the tip of the core member 1102 shown in FIG. 11 relative to the display 230 shown in FIG. 2 on the e-paper tablet 110). Thus, in one embodiment of the invention, as the tilt angle θ increases, the shading area proportionately increases on the display. One could similarly imagine a slightly different arrangement of the components such that as the tilt angle θ decreased, the area of marking area proportionately increased, e.g., in inverse proportion. In both embodiments, the area of marking still has a direct correlation to the tilt angle θ , e.g., as the tilt angle changes, the area of marking changes.

This tilt angle determination would operate in a similar manner for an erasure function. Embodiments of the invention allow for the derivation of the tilt angle of the tail eraser by employing an electronically calculated method in the input digitizer 220 similar to that employed conventionally by the input digitizer 220 for determining the tilt of the marker tip and the eraser function, when present. Such a calculation requires that the pen-stylus provide additional data/information to the input digitizer 220.

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In some embodiments, the pen-stylus integrated circuits (e.g., included in the PCBA **1105**) may not have a sufficient number of antenna signal lines for driving two transmitters in both the active pen tip and tail eraser antenna system. Conventional active pens have two antenna signal lines for the active pen-stylus tip and one antenna signal line for a tail eraser. Thus, no more than three antenna signal lines. With this conventional configuration, the input digitizer **220** (or comparable hardware) cannot derive the tilt information for the tail eraser. This problem has been solved in a manner that allows a conventional active pen configuration to drive four antennas rather than the conventional three antennas, thus enabling enhanced erasure capabilities. Further details about structures and functions for solving this problem may be found in U.S. application Ser. No. 18/208,280 to Gaute Nordby et al., entitled "Active Pen-Stylus Precise Eraser," which is incorporated by reference herein.

FIG. **14** illustrates a cross section of the forward portion of an active pen-stylus **1400** that a user may employ for tasks such as drawing a line on the display of a device (e.g., the display **303** of the e-paper tablet **300** shown in FIG. **3**), according to an embodiment of the invention. The active pen-stylus **1400** includes a core (e.g., the core member **1102** shown in FIG. **11**) comprised of components such as a first antenna **1407** and a second antenna **1425**. The active pen-stylus **1400** also includes a force sensor (e.g., the force sensing system **1104** shown in FIG. **11**) comprised of components such as a writing shaft **1413**, a first spring coil **1431**, a second spring coil **1429**, and a force sensor **1423**.

The active pen-stylus **1400** provides high accuracy for various drawing tasks due to interactions between the first antenna **1407** and the second antenna **1425**, in part because of the placement of the first antenna **1407** and the second antenna **1425** with respect to each other and with respect to the active pen-stylus **1400** itself. As shown in FIG. **14**, the second antenna **1425** surrounds the first antenna **1407**, but the second antenna **1425** and the first antenna **1407** do not overlap horizontally in the active pen-stylus **1400** and are further separated from each other by the insulator **1405**.

Each antenna **1407**, **1425** may send a clear signal to the e-paper tablet **110**. This arrangement is particularly helpful for the antenna **1425** since the antenna **1407** is often in physical contact with the display of the e-paper tablet **110**. When the pen-stylus **1400** is not in contact (e.g., out of range) with the tablet (e.g., the e-paper tablet **110**), antennas on the tablet periodically send a beacon signal. This beacon signal is meant for the pen-stylus **1400** to detect that it is within range of the tablet. In a time following the beacon signal, the tablet typically keeps an open time slot for the pen-stylus **1400** to acknowledge the beacon and respond to the tablet. This acknowledgement initiates the two-way communication between the pen-stylus **1400** and the tablet (e.g., the e-paper tablet **110**). As an ordinary artisan will recognize, the "beacon" from the tablet is a form of an uplink to the pen-stylus **1400**, and the communication from the pen-stylus **1400** to the tablet is a form of a downlink.

The active pen-stylus **1400** also offers advanced force (e.g., pressure) sensing, e.g., a low activation force, high maximum force, and a high dynamic range. To translate the detecting force imparted by the pen-stylus **1400** against the display of the e-paper tablet **110**, the writing shaft **1413** is designed to move towards and/or away from a force sensor **1423**, which registers activity between a replaceable marker tip **1403** and the display (e.g., the display of the e-paper tablet **110**), including the pressure applied by the user of the pen-stylus **1400** to the display. Among other things, a bottom bracket **1419** prevents the writing shaft **1413** from rotating

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about its center axis. As previously discussed, the writing shaft **1413** is designed to move towards and/or away from a force sensor **1423**, which registers activity between the replaceable marker tip **1403** and the display (e.g., the display of the e-paper tablet **110**), including the force (e.g., pressure) applied by the user of the pen-stylus **1400** to the display. The active pen-stylus **1400** has been designed to have a minimum set of components, according to an embodiment of the invention. Further details about structures and functions of active pen-stylus writing systems can be found in U.S. application Ser. No. 18/779,164, filed on Jul. 22, 2024 entitled "Marker Writing System" and given reference number "21149403," which is incorporated by reference herein.

The active pen-stylus **1400** offers a variety of device safety features for its operational protection and long service life. Among other things, the writing shaft **1413** may be fitted with an impact shock **1417** that is designed to receive heavy forces imparted to the pen-stylus **1400** (e.g., the force arising from being dropped onto a hard surface) so that an upper limit is set for the forces transmitted to the force sensor **1423**.

The replaceable marker tip **1403** has also been designed to leave a marker tip clearance **1411** between itself and the marker body **1401**, according to an embodiment of the invention. The marker tip clearance **1411** may also be helpful in receiving certain shock forces to the pen-stylus **1400** that might otherwise be received by components inside the pen-stylus **1400**. In the event of a severe shock, the replaceable marker tip **1403** may also function as a sacrificial element that may even absorb otherwise destructive energy. In situations where the replaceable marker tip **1403** has absorbed so much destructive energy that it can no longer function, then the user of the pen-stylus **1400** may simply need to add a new replaceable marker tip **1403** to an otherwise workable pen-stylus **1400**.

Further details about structures and functions of active pen-stylus safety systems can be found in U.S. application Ser. No. 18/779,158, filed on Jul. 22, 2024 entitled "Marker Protection System," and given reference number "21149402," which is incorporated by reference herein.

As described further below, the replaceable marker tip **1403** is a removable and replaceable component. Users of the pen-stylus **1400** may wish to have a readily accessible collection of new replaceable marker tips in a specialized holder designed to hold marker tips securely and facilitate their attachment to the pen-stylus **1400**. Further details about structures supporting a marker tip repository can be found in U.S. application Ser. No. 18/779,170, filed on Jul. 22, 2024 entitled "Captive Object Flexure Mechanism," and given reference number "21149404," which is incorporated by reference herein.

The pen-stylus **1400** may include other components, such as a writing system spring **1421** which in cooperation with the force sensor **1423** may present to the user of the pen-stylus **1400** the feeling of writing on a stack of paper. Further details about structures and functions of active pen-stylus paper emulation can be found in U.S. application Ser. No. 18/779,154, filed on Jul. 22, 2024 entitled "Advanced Paper Emulation," and given reference number "21149401," which is incorporated by reference herein. The Replaceable Conductive Marker Tip

The first antenna **1407** forms a portion of a removable marker tip **1403** that also includes an insulator **1405**. The marker tip **1403** removably attaches to the writing shaft **1413** via a series of crush ribs **1409** located in an interior cavity on the marker tip **1403**, according to an embodiment of the invention.

FIG. 15 illustrates a replaceable marker tip 1403 for the pen-stylus 1400 that provides a good writing feeling (e.g., paper emulation) for users of the e-paper tablet 110 while also providing a high quality antenna function, specifically serving as the main antenna for the pen-stylus 1400, according to an embodiment of the invention.

The marker tip 1403 creates a high-quality writing feeling (e.g., paper emulation) by providing a consumable element in the interface between the pen-stylus 1400 and the e-paper tablet 110. Many of the marker tips in the prior art are based on conventional technology. In contrast, the marker tip 1403 has been customized for operation with devices such as the e-paper tablet 110 that aim to replicate the feel of paper, enabled in part, by allowing the marker tip 1403 to be consumed slowly during use.

While possibly more frequent in passive than active pens, only a few marker tips have previously been designed as consumable parts. However, since the main objective of the e-paper tablet 110 is to emulate paper, specifically the feel of paper texture, special care must be undertaken to achieve this goal. The emulation of a paper feeling often requires that some component (e.g., the tip of the pen-stylus 1400) should be amenable to wear (e.g., consumption) as a tradeoff for attaining the feel of paper for users. When such a tradeoff is in play, then it is generally easier and cheaper to replace a component like a marker tip than another component like a display surface. Of course, it might be possible to construct a marker tip of novel materials that aids in emulating paper but does not wear; however, such materials are not readily available on the market at present.

As shown in FIG. 16, the marker tip 1403 comprises a core antenna 1407 and an insulating material 1405, according to an embodiment of the invention. The core antenna 1407 serves both as an antenna for the pen-stylus 1400 and has a specific and defined geometric shape that has been designed to provide good signaling properties and which also provides a tactile feeling during operation that emulates a paper writing feeling for users, especially in combination with a specialized texture on the display of the e-paper tablet 110.

The ideal antenna for a marker comprises a spherical shape. Accordingly, the front portion of the core antenna 1407 comprises a hemisphere. The shape of the core antenna 1407 replicates a sphere well at most angles, degrading slightly at shallow angles. Thus, the signal remains consistent even when the user has tilted the pen-stylus 1400 at a shallow angle.

As mentioned, the core antenna 1407, serves as antenna-1 for the pen-stylus 1400. Embodiments of the pen-stylus 1400 may include multiple antennas, as discussed, especially with respect to FIGS. 13A-13B. These antennas may be combinations of transceivers, transmitters, and receivers, e.g., one transceiver and one transmitter. Antenna 1 or Antenna 2 may be used as transceivers in such antenna systems which may be programmable by the electronics in the pen-stylus 1400, e.g., the electronics of the PCBA 1105. To select which antenna that also acts as a receiver (will then be a transceiver), one needs to take into account factors such as the receiver capabilities for that antenna, the signal to noise ratio of the system, processing capabilities of the active-pen ASIC and MCU etc. Antenna-1 comprises a transmitter, according to an embodiment of the invention. Antenna-2 (e.g., antenna 1425 shown in FIG. 14)) typically comprises a transceiver, according to an embodiment of the invention. The Antenna-1 is the main antenna for the pen-stylus 1400, and also serves as the main contact with the display of the e-paper tablet 110. Antenna-2 provides a

secondary signal that may be used to determine the marker's orientation, as discussed above with respect to FIGS. 13A-13B, according to an embodiment of the invention.

The marker tip's core antenna 1407 comprises an antenna that is composed of a conductive material (e.g., carbon fiber). The conductive carbon fiber material of the core antenna 1407 may comprise a carbon fiber percentage. For example, the core antenna 1407 may comprise a polymeric binder (e.g., polyamide, "nylon") filled with a sufficient amount of carbon fiber, carbon nanotubes (CNT), carbon black powder, carbon nanofibers (CNF), graphite, conductive mica, conductive mica (powder), carbon fiber (chopped, milled), multi-wall carbon nanotubes (MWCNT), and/or graphene nanotubes (GNT).

Voltages applied by the core antenna 1407 should be as high as needed to create a strong signal to noise ratio. Higher voltages provide a clearer signal but too high voltages may create difficulties. Among other things, the specific voltages applied depend on the integrated circuit (e.g., the PCBA 1105 shown in FIG. 11) employed in the pen-stylus 1400.

The insulating material 1405 for the marker tip 1403 comprises an insulator, such as glass-filled nylon. The insulating material 1405 improves the mechanical properties of the marker tip 1403, such as increasing the toughness of the marker tip 1403. The stiffness of the marker tip 1403 may be helpful in replicating the feel of a pencil for users. The insulating material 1405 should keep the capacitive coupling between Antenna-1 and Antenna-2 as small as possible. If there is capacitive coupling between the two antennas, it makes the signal processing on the tablet (e.g., the e-paper tablet 110) difficult as the signal will couple through the other antenna obfuscating the signal leading to less precise position or tilt calculation. It is likewise important that the electro-mechanical design for the pen-stylus 1400 is done in such a way that capacitive coupling between the signals are avoided as much as possible.

The marker tip 1403 has been designed to be able to push onto the pen-stylus 1400, specifically the writing shaft 1413. The ease of replacing the marker tip 1403 on the pen-stylus 1400 arises from its nature as a replaceable part, as mentioned above.

The marker tip 1403 is held on by force to the writing shaft 1413. The marker tip 1403 has also been designed not to rotate on the writing shaft 1413. Among other things, if the marker tip 1403 rotated, then the signal between the pen-stylus 1400 and the display of the e-paper tablet 110 would cause instability. The writing shaft 1413 does not rotate by locking it to bottom bracket 1419. In addition, interference between the writing shaft 1413 and the marker tip 1403 is carefully designed so that torque applied when writing and/or using the marker tip 1403 does not exceed the interference force, according to an embodiment of the invention.

As shown in FIGS. 17-19, crush ribs 1409 are critical for holding the core antenna 1407 onto the writing shaft 1413. The marker tip 1403 connects to the pen-stylus 1400, specifically, the writing shaft 1413 by employing the crush ribs 1409 in the core antenna 1407 that presses the core antenna 1407 around and on the inside of the writing shaft 1413 rather than on the outside of the writing shaft 1413. Thus, the marker tip 1403 can be said to go onto the writing shaft 1413 rather than into it, according to an embodiment of the invention. In an alternative embodiment of the invention, the marker tip 1403 could be designed to go into the writing shaft 1413, which as discussed above is the main vessel for transferring forces to the force sensor 1423.

As mentioned, the material composition of the marker tip **1403** will wear (e.g., be slowly consumed) and will need to be replaced from time to time, according to an embodiment of the invention. More specifically, the core antenna **1407**, as well as the insulating material **1405**, will wear and will need to be replaced. The operation lifetime for a given core antenna **1407** depends on user behavior and user preference. The core antenna **1407** might last longer than six months for some users, while it might last just one month for other users. As mentioned, the lifetime for the marker tip **1403** depends on a variety of factors related to its use, such as the force applied while using the pen-stylus **1400**, how much and how often the user tilts the pen-stylus **1400**, and how often the user rotates the pen-stylus **1400**, along with the user's own personal sensitivity. Put another way, some users may find the marker tip **1403** to be perfectly fine when it is nearly worn down to the marker tip clearance **1411**—even at that point, some portion of the core antenna **1407** still remains—conversely, other users may find the fidelity of the marker tip **1403** not suiting their needs even before the wear on the marker tip **1403** has abraded the core antenna **1407** down to the beginning of the insulating material **1405**.

As mentioned, the removable marker tip **1403** should be removable by hand and require no specialized tool for its removal, according to an embodiment of the invention. Conversely, the marker tip **1403** should not slip or fall out of the pen-stylus **1400**. Likewise, the marker tip **1403** should not spin and/or cause wear for other parts of the pen-stylus **1400**.

Depending on the specific design and materials employed, the force needed to remove the marker tip **1403** applied by a user (e.g., by hand and without a tool) may be set in a range between 2 Newtons to just below 8 Newtons, according to an embodiment of the invention. Experimentation has shown that the axial force to pull off marker tip **1403** should be at least 2 Newtons to prevent the marker tip **1403** from being able to spin on its axis. The preference is 2 Newtons but could be nearly 8 Newtons, according to an embodiment of the invention.

Finally, a pen-stylus **1400** safety feature involves the removable marker tip **1403**, according to an embodiment of the invention. Specifically, a positive stop **1427** shown in FIG. **14** prevents the writing shaft **1413** from being removed from the pen-stylus **1400**. The positive stop **1427** may be included in a writing shaft captivator **1415**. As discussed, the removable marker tip **1403** is designed to be removed by the user by hand and without the need for a tool. During this replacement process, the writing shaft **1413** may possibly experience some forces that might otherwise disrupt its location without the presence of the positive stop **1427**; put another way, when the force applied by a user in replacing the marker tip **1403** exceeds the 2N to 8N range specified above, then the marker tip **1403** comes off the pen-stylus **1400**.

The marker tip **1403** may be inexpensive to manufacture and is designed to be interchangeable with replacement marker tips, according to an embodiment of the invention. While inexpensive, as a customized device, the marker tip **1403** requires care in its manufacture. Accordingly, the marker tip **1403** may be difficult to manufacture if caution is not practiced.

As shown by comparing FIG. **15** and FIG. **16**, the marker tip **1403** comprises an overmolded design, according to an embodiment of the invention. The core antenna **1407** is overmolded with the insulating material **1405** to create a more appealing visual geometry and to function as an

insulator for the constraining component (e.g., the writing system portion of the pen-stylus **1400**).

The core antenna **1407** should provide a continuous signal to the writing shaft **1413** for proper operation. In other words, the core antenna **1407** should not have interior breakages that weaken the signal. Thus, the core antenna **1407** should be conductive throughout its entire length, otherwise the marker tip **1403** may insulate or block the transport of signals related to the user's activities on the display of the e-paper tablet **110**. In the general case, if resistance increases, then signaling becomes an issue.

The inscribed diameter of the marker tip **1403** provides 20 microns of tolerance, according to an embodiment of the invention. Otherwise, the inscribed diameter across the crush ribs **1409** of the marker tip **1403** might be too loose or too tight on the outer diameter of the writing shaft **1413**. The volume of material in the crush ribs **1409** dictates the pull off force, and achieving a good force range may be attained by having tight manufacturing tolerances, according to an embodiment of the invention.

Finally, the core antenna **1407** and the insulating material **1405** may be color matched in some embodiments to provide a more aesthetically pleasing appearance.

ADDITIONAL CONSIDERATIONS

As discussed below, these pen-styluses can be equipped with a tail eraser allowing the user to erase content from the display of the computing device, e.g., the e-paper tablet **110**. An active pen includes electronics components which enables the active pen to send and receive signals from the computing device.

This disclosed configuration provides additional precision and options for users as they go about erasing portions of drawings on an e-paper tablet. This should improve the efficiency of users interacting with e-paper tablets while also enabling them with more precise functional capabilities.

It is to be understood that the figures and descriptions of the present disclosure have been simplified to illustrate elements that are relevant for a clear understanding of the present disclosure, while eliminating, for the purpose of clarity, many other elements found in a typical system. Those of ordinary skill in the art may recognize that other elements and/or steps are desirable and/or required in implementing the present disclosure. However, because such elements and steps are well known in the art, and because they do not facilitate a better understanding of the present disclosure, a discussion of such elements and steps is not provided herein. The disclosure herein is directed to all such variations and modifications to such elements and methods known to those skilled in the art.

Some portions of above description describe the embodiments in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as engines, without loss of generality. The described operations and their associated engines may be embodied in software, firmware, hardware, or any combinations thereof.

As used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, struc-

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ture, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the invention. This description should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise. While particular embodiments and applications have been illustrated and described, it is to be understood that the disclosed embodiments are not limited to the precise construction and components disclosed herein. Various modifications, changes and variations, which will be apparent to those skilled in the art, may be made in the arrangement, operation and details of the method and apparatus disclosed herein without departing from the spirit and scope defined in the appended claims.

The invention claimed is:

1. An active pen-stylus, comprising:

- a removable marker tip having a first antenna and an insulator;
- a writing shaft adapted to receive and hold the removable marker tip;
- a force sensor that receives physical forces imparted to the writing shaft by the removable marker tip when the removable marker tip engages with a display on a tablet device and translates the physical force received into an electronic signal, wherein the electronic signal is transmitted back to the tablet device; and

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a second antenna, wherein the second antenna surrounds the first antenna but has a horizontal placement in the active pen-style that avoids overlap with the first antenna,

wherein signals received by the tablet device from the first antenna and the second antenna enable the tablet device to calculate an orientation of the active pen-stylus with respect to the display of the tablet device.

2. The active pen-stylus of claim 1 wherein the first antenna comprises a transmitter and the second antenna comprises a transceiver that receives signals from tablet device.

3. The active pen-stylus of claim 1 wherein the removable marker tip further comprises a posterior opening having crush ribs that fit onto the writing shaft and secure the removable marker tip to the writing shaft.

4. The active pen-stylus of claim 3 wherein the crush ribs are aligned to avoid delivering rotational forces to the writing shaft.

5. The active pen-stylus of claim 1 wherein the first antenna of the removable marker tip comprises a first consumable material and the insulator comprises a second consumable material wherein the first consumable material and the second consumable material emulate for users of the active pen-stylus sensations similar to applying one of a pen and pencil to paper when using the active pen-stylus to draw on the display of the tablet device.

6. The active pen-stylus of claim 5 wherein the first consumable material of the first antenna comprises a conductive carbon fiber material.

7. The active pen-stylus of claim 5 wherein the first consumable material of the first antenna comprises a polymeric binder filled with at least one of a carbon fiber, carbon nanotubes (CNT), carbon black powder, carbon nanofibers (CNF), graphite, conductive mica, conductive mica (powder), carbon fiber (chopped, milled), multi-wall carbon nanotubes (MWCNT), and graphene nanotubes (GNT).

8. The active pen-stylus of claim 5 wherein the second consumable material of the insulator comprises a glass-filled nylon.

9. The active pen-stylus of claim 1 wherein the removable marker tip is hand-removable from the writing shaft.

10. The active pen-stylus of claim 1 wherein the first antenna is overmolded with the insulator.

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